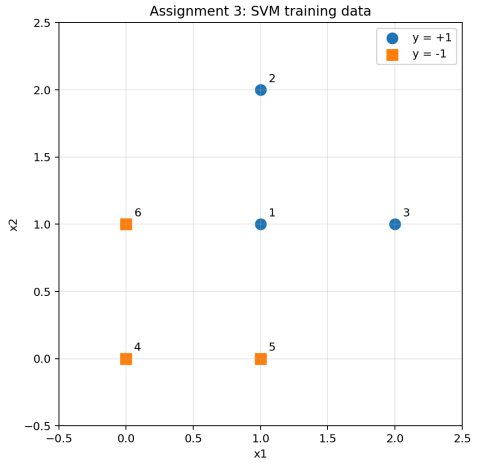
Question 1

(a) Plot and support vectors



Training data.

Support vectors: Points 1 (1, 1), 5 (1, 0), and 6 (0, 1).

The positive points lie on $x_1 + x_2 = 2$ (Point 1) and $x_1 + x_2 = 3$ (Points 2 and 3). The negative points lie on $x_1 + x_2 = 1$ (Points 5 and 6) and $x_1 + x_2 = 0$ (Point 4).

The closest parallel lines containing points from the two classes are $x_1 + x_2 = 1$ and $x_1 + x_2 = 2$. Points 1, 5, and 6 lie on these supporting lines and determine the maximum-margin separator. Points 2, 3, and 4 lie farther from the separating boundary and are not support vectors.

(b) Boundary and margin

Boundary. For a hard-margin SVM, the constraints are

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1,$$

with equality at the support vectors. Substituting Points 1, 5, and 6 gives

$$\begin{aligned} w_1 + w_2 + b &= 1, \\ w_1 + b &= -1, \\ w_2 + b &= -1. \end{aligned}$$

Subtracting the third equation from the second gives $w_1 = w_2$. Subtracting the second equation from the first gives $w_2 = 2$. Therefore, $w_1 = 2$ and $b = -3$.

The resulting maximum-margin separating boundary is

$$\boxed{2x_1 + 2x_2 - 3 = 0},$$

where $\mathbf{w} = (2, 2)$ and $b = -3$.

For Points 1 through 6, the values of $y_i(\mathbf{w}^\top \mathbf{x}_i + b)$ are $1, 3, 3, 3, 1, 1$, respectively. All hard-margin constraints are satisfied.

Geometric margin. The geometric margin is the perpendicular distance from the decision boundary to the closest training points:

$$\begin{aligned} \gamma &= \frac{1}{\|\mathbf{w}\|} \\ &= \frac{1}{\sqrt{2^2 + 2^2}} \\ &= \boxed{\frac{1}{2\sqrt{2}} \approx 0.3536}. \end{aligned}$$

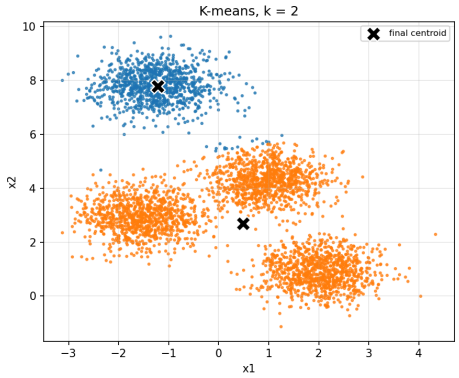
Question 2

(a) Update steps

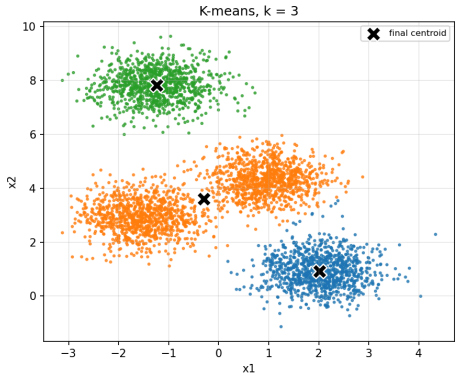
`update_assignments()` computes the squared Euclidean distance from each point to every centroid and assigns the point to the nearest centroid. If multiple centroids have the same distance, `np.argmin` selects the centroid with the lowest index.

`update_centroids()` updates each nonempty cluster's centroid to the mean of its assigned points. If a cluster has no assigned points, its previous centroid is retained.

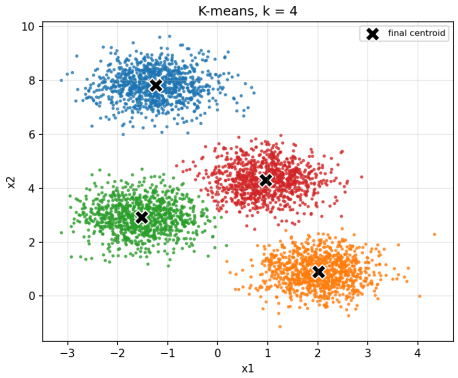
(b) Results for k = 2, 3, 4, 6, 10



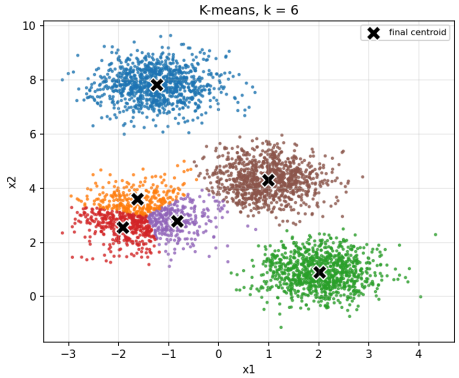
k = 2



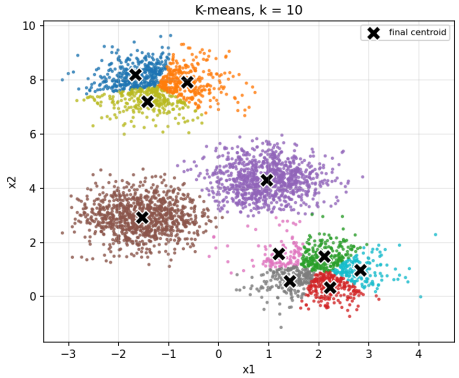
k = 3



k = 4



k = 6



k = 10

Discussion.

As k increases, K-means divides the data into more, smaller clusters. At $k = 2$, three generating groups are merged into one cluster, while the upper group remains separate. At $k = 3$, the two middle groups remain merged. At $k = 4$, the clusters closely match the four generating groups, with centroids near their centers. At $k = 6$, the left-middle group is split into three clusters. At $k = 10$, the upper group is split into three clusters and the lower-right group into five. Thus, using fewer than four clusters merges generating groups, while using more than four splits existing groups into smaller clusters.

Question 3

(a) Noisy-label baseline

Mean clean-test accuracy over 20 trials: 0.82795.

(b) K-NN label repair

Method	Mean clean-test accuracy (20 trials)
Baseline (noisy labels)	0.82795
K-NN repaired labels	0.86440
Difference (K-NN − baseline)	**+0.03645**

Discussion. Across 20 trials, 5-NN label repair increased the SVM's mean clean-test accuracy from 82.795% to 86.440%, a gain of 3.645 percentage points. Neighbor voting can correct isolated label flips, but it can also change correct labels where the two classes overlap.

AI Acknowledgment

ChatGPT and Claude were used to help interpret the requirements and check/polish my answers.

https://chatgpt.com/share/6aeacd4c-05b0-83e8-ae3f-90bc8ffa6862

https://claude.ai/share/e5aabe1c-f815-4c0a-854b-77e30a3a13fd